

California Housing Dataset – EDA Summary

1. Dataset Overview:

The California Housing dataset contains information about houses and different characteristics of housing areas in California. It contains 20,640 records and 8 input features. The main features include median income, house age, average number of rooms, average number of bedrooms, population, average occupancy, latitude, and longitude. The target value represents the median house value of the area.

There were no missing values in the dataset, so no missing-value treatment was required.

2. Important Data-Quality Problems:

The first important issue is the target value limit. The maximum target value is 5.00001, and many values are concentrated near this upper limit. This suggests that very high house values were limited to a maximum value in the dataset. Because of this, a model may not properly predict the actual values of very expensive areas.

The second important issue is the presence of extreme values in some features. Features such as AveRooms, AveBedrms, and AveOccup contain some unusually large values. These values can have a strong effect on the analysis and should be checked carefully. A transformation such as a log transformation could be considered to reduce the effect of these extreme values.

Another important observation is that houses in nearby geographical locations can have similar characteristics. This means that the records are not always completely independent from each other.

3. Three Predictors I Would Use First:

MedInc:

I would use MedInc (median income) first because it has the strongest relationship with the target in our analysis. Its correlation with the target was approximately 0.688, which was much higher than the other features.

AvgRooms:

I would use AvgRooms (average number of rooms) as the second predictor. It has a positive relationship with the target and can provide information about the size and type of housing in an area.

HouseAge:

I would use HouseAge (average house age) as the third predictor. It has a positive relationship with the target and provides useful information about the age of houses in the area.

Conclusion:

Overall, MedInc, AvgRooms, and HouseAge would be my first choices for predicting house values. MedInc is especially important because it showed the strongest relationship with the target. Before building a prediction model, I would also pay attention to the extreme values and the upper limit on the target values. Geographic location should also be considered because nearby areas can have similar house values.